

Enter *all* of the high-priority design requirements you listed in Element C into the table below.

Some of the requirements are that the device must be portable. This device also has to be cost-friendly, has to provide some type of shade, and has some type of mobility. This device also needs to be capable of operating by itself and be environmentally safe. It should also be able to withstand at least 10 pounds. The platform should be a flat-surface so the plant doesn't tip over when moving to a place with shade after receiving the necessary amount of sunlight. It should be battery powered for 1 week without the need for recharge.

Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.

As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.

Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:

how the test will be set up-The test will be whenever we get access to a plant and have to see how the plant health is after a certain amount of time.We have to see how the program and solar panel works.

what needs to happen during the test-The plant needs to be in the prototype for a certain amount of time and it needs to track the data like temperature and the amount of sunlight the plant is getting.

test cases (if applicable)-

number of trials of the test expected (aim for 7 or more trials)

the expected range of results that could be recorded, including units

the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds

any goals your team has set beyond the passing result

Passing-The plant is healthy and not heat damaged after a few days in the device.The device is moveable.

Fail-Design doesn't meet requirements and kills the plant.

requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
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1	The prototype must remain intact when held upside down.	1) Hold the prototype in a hand using the grip in this photo.	a.	Parts of the prototype may:	Fail: one or more parts of the prototype tears or detaches
		<insert grip photo>	Rotate around x-axis (10 trials)	– remain unchanged	
		2) Take 3 seconds to slowly rotate the prototype around the indicated axis into an upside down position with a smooth motion.	b.	– deform	Pass: zero parts of the prototype tear or detach
		<insert photo with annotation of axes of rotation>	Rotate around y-axis (10 trials)	– shift position	
		3) Note any observations about prototype integrity.	c.	– tear	Goal: all parts of the prototype remain unchanged when the prototype is returned to its original position
		4) Take 3 seconds to rotate the prototype in reverse, back to its original position, using a smooth motion	Rotate around z-axis (10 trials)	– detach	

2	mass of prototype < 3kg	1) Tare a digital scale.	test cases:	0kg<reading <10kg	Fail: mean of the trials $\geq$ 3kg
		2) Place the prototype securely on the scale.	N/A		Pass: mean of the trials < 3kg
		3) Wait for the reading on the scale to stabilize before recording the value.	# trials: 10		Goal: mean of the trials < 2kg



